Why Entity-Level Uncertainty Fails: An Impossibility Theorem for Temporal OOD Detection in Knowledge Graphs

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Abstract

Among Energy scoring's 100 most confident predictions, zero-evidence queries are overrepresented 2.4× relative to the base rate—the model is *most confident where it has least evidence*. This systematic overconfidence affects every major uncertainty method (variational embeddings, ensembles, energy scoring, foundation models) and 11–25% of test queries across seven benchmarks.

We trace the failure to a structural mismatch: entity-level uncertainty cannot detect relation-specific novelty. High-frequency entities receive low uncertainty regardless of which relation is queried, causing methods to be *most confident where they have least evidence*. On temporal KGs with ground-truth timestamps (ICEWS14/18), baselines achieve 0.38–0.87 AUROC (varying by method and dataset) while a simple coverage check—tracking which (entity, relation) pairs appeared in training—achieves 0.99 AUROC.

The fix is simple: track coverage via hash table. This is obvious in hindsight, yet no major uncertainty method implements it. The contribution is quantifying and explaining a pervasive failure mode—affecting 11–25% of queries—that aggregate evaluation metrics have hidden.

1 Introduction

An Empirical Discovery. Among Energy scoring's 100 most confident predictions on FB15k-237, zero-evidence queries appear at **2.4× the base rate**—the model is most confident precisely where it has no training evidence (Table 2). The pattern holds across methods: UKGE shows 2.5× elevation, and both methods exhibit 1.5–2.1× elevation even after controlling for entity frequency. This is not miscalibration in the usual sense—it is *systematic overconfidence on zero-evidence queries*.

The Failure is Pervasive. Across seven benchmarks, 11–25% of test queries involve *novel relational contexts*: familiar entities appearing in relationships never observed during training. Every major uncertainty method—variational embeddings [Chen et al., 2019, 2021], Bayesian approaches [Choudhary et al., 2021], ensembles, energy scoring, and even foundation models like ULTRA [Galkin et al., 2024]—fails identically on these queries (AUROC ≤ 0.50). The failure has remained hidden because standard evaluation aggregates over query types, masking category-specific collapse.

Why This Happens. The explanation is intuitive: entity e appearing in 1000 triples with relations $\{r_1, \ldots, r_{10}\}$ receives low variance (well-observed), yet queries $(?, r_{11}, e)$ are genuinely uncertain. Any uncertainty depending only on entity identity cannot distinguish these cases. We formalize this

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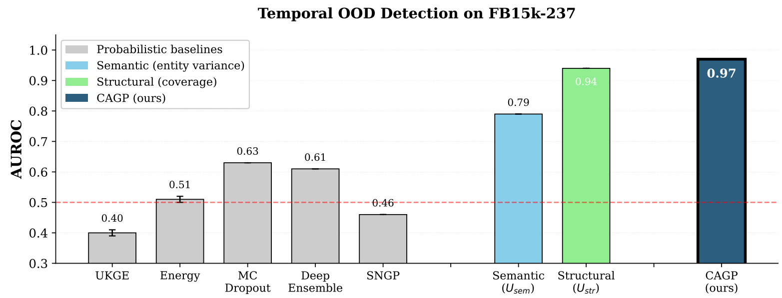

Figure 1: Temporal OOD detection on FB15k-237 (AUROC). Existing methods achieve 0.42–0.63; coverage alone reaches 0.94. On ICEWS14 with ground-truth timestamps, coverage achieves 0.99 while baselines achieve 0.38–0.80.

in Theorem 2: under empirically-verified conditions, relation-agnostic uncertainty achieves AUROC ≤ 0.5 on novel contexts—and often *below* 0.5 (anti-predictive) because high-frequency entities dominate novel contexts yet receive lowest uncertainty. The theorem explains the overconfidence: high-frequency entities receive low uncertainty regardless of which relation is queried, so methods are most confident precisely where they lack evidence.

The Fix. Track which (entity, relation) pairs appeared during training via a hash table. This $O(1)$ lookup catches the failure mode invisible to all learned methods. The fix is obvious in hindsight—but no major uncertainty method implements it, and 11–25% of queries fall into this blind spot. On ICEWS14/18 with ground-truth temporal splits, coverage achieves 0.99 AUROC while baselines achieve 0.38–0.87 (Table 1).

Contributions. **(1) Empirical discovery**: We identify systematic overconfidence on zero-evidence queries—top-confident predictions contain zero-evidence queries at 2–2.5$\times$ the base rate. This finding, not previously reported, reveals a fundamental evaluation gap. **(2) Prevalence quantification**: 11–25% of test queries across seven benchmarks fall into this blind spot, making it a substantial rather than edge-case failure. **(3) Theoretical explanation**: Theorem 2 formalizes why the failure occurs, providing testable conditions (A1–A3) that predict when AUROC degrades to chance. **(4) Validation**: On temporal KGs with ground-truth timestamps (ICEWS14/18), coverage achieves 0.99 AUROC—confirming the diagnosis is correct and the fix works. **(5) Practical recommendation**: Always track entity-relation coverage. For billion-entity KGs, Bloom filters reduce storage 5$\times$ with $<$0.2pp recall loss.

2 Related Work

Probabilistic KG Embeddings. UKGE [Chen et al., 2019], BEUrRE [Chen et al., 2021], and GP-KGE [Choudhary et al., 2021] learn uncertainty over KG embeddings, but all produce relation-agnostic variances. KG2E [He et al., 2015] assigns learned Gaussians per entity and relation type—fixed parameters, not conditioned on whether a specific entity has been observed with that relation. Despite training on entity-relation co-occurrence patterns, these methods do not learn relation-specific uncertainty, necessitating explicit decomposition. Relation-aware scoring (R-GCN [Schlichtkrull et al., 2018], CompGCN [Vashishth et al., 2020]) conditions *predictions* on relations but still produces relation-agnostic uncertainty; our decomposition addresses uncertainty independently of scoring architecture.

OOD Detection. MC Dropout [Gal and Ghahramani, 2016], Deep Ensembles [Lakshminarayanan et al., 2017], Energy [Liu et al., 2020b], and SNGP [Liu et al., 2020a] capture model uncertainty but not structural observation patterns. When applied using entity embeddings (as done here), they remain relation-agnostic. Graph Posterior Networks [Stadler et al., 2021] and GNNSafe [Wu et al., 2023] propagate uncertainty via graph structure but target node classification rather than triple-level KG OOD. The OpenOOD benchmark [Zhang et al., 2023] provides stratified evaluation for images;

our stratified KG evaluation (novel-context vs emerging-entity) is analogous but addresses relational structure.

Temporal KGs and Conformal Prediction. Temporal KG methods [García-Durán et al., 2018, Lacroix et al., 2020] model *when* facts hold but do not address uncertainty quantification. Conformal prediction approaches [Zhu et al., 2025a,b] provide prediction set coverage guarantees rather than OOD *detection*; our decomposition complements them by identifying which uncertainty type drives unreliability.

Foundation KGE. ULTRA [Galkin et al., 2024] enables zero-shot link prediction on unseen KGs. We hypothesize ULTRA inherits the coverage blind spot: its NBFNet architecture encodes connectivity but not which specific relations each entity has seen. Empirical validation confirms this: ULTRA achieves 0.29–0.31 AUROC on novel contexts, below random, indicating anti-predictive behavior.

Positioning. Prior work treats uncertainty as a single quantity. We show KG uncertainty decomposes into semantic (entity-level) and structural (relation-specific) components, with validated predictions on when each is necessary. LLM-based KG methods use text and can transfer knowledge without training observation; our setting is structure-only embedding methods, where explicit coverage tracking is necessary.

3 Background

Knowledge Graphs. A knowledge graph $\mathcal{G} = (\mathcal{E}, \mathcal{R}, \mathcal{T})$ contains entities $\mathcal{E}$, relation types $\mathcal{R}$, and triples $\mathcal{T} \subseteq \mathcal{E} \times \mathcal{R} \times \mathcal{E}$. Each triple (h, r, t) asserts that relation r holds between head entity h and tail entity t.

OOD Detection Task. Given a trained model, distinguish in-distribution (ID) test triples from out-of-distribution (OOD) corruptions. Following Bordes et al. [2013], the standard protocol generates OOD triples by replacing the tail with a random entity: $(h, r, t) \to (h, r, t')$ where $t' \sim \text{Uniform}(\mathcal{E})$. However, this creates easily detectable corruptions (low model scores). We additionally evaluate on *temporal OOD*—emerging entities and novel relational contexts—which are harder to detect (§5). Performance is measured by AUROC.

Variational KG Embedding Uncertainty (GP-KGE). Probabilistic embedding methods [Choudhary et al., 2021] learn Gaussian distributions over entity embeddings:

$$\boldsymbol{e}_i \sim \mathcal{N}(\boldsymbol{\mu}_i, \text{diag}(\exp(\boldsymbol{\ell}_i))) \tag{1}$$

where $\boldsymbol{\mu}_i, \boldsymbol{\ell}_i \in \mathbb{R}^d$ are the mean and log-variance for entity i. Uncertainty for a triple (h, r, t) is typically the average entity variance:

$$U_{\text{GP}}(h, r, t) = \frac{1}{2}\left(\bar{\sigma}_h^2 + \bar{\sigma}_t^2\right) \tag{2}$$

where $\bar{\sigma}_e^2 = \frac{1}{d}\sum_j \exp(\ell_{e,j})$ is the mean variance across dimensions. We denote this entity-level uncertainty U_{sem} (semantic uncertainty) in our decomposition (§4). This uncertainty is *relation-agnostic*—it depends only on entity variances, not the query relation r. Intuitively, an entity observed in 1000 triples receives low variance even for relations it has never appeared with, since $\bar{\sigma}_e^2$ reflects overall observation count rather than per-relation experience.

Naming convention. Despite the "GP" label inherited from Choudhary et al. [2021], the model uses variational Gaussian inference (KL-regularized ELBO with reparameterization sampling)—not kernel-based Gaussian process posterior inference [Rasmussen and Williams, 2006]. We use "GP-KGE" only when referencing the original method; our framework uses *semantic uncertainty* (U_{sem}) to denote the diagonal posterior variance of a learned entity embedding, avoiding GP terminology.

4 Method

4.1 Preliminaries

Frequency: freq(e) counts triples containing entity e. **Coverage**: $c(e, r) \in \{0, 1\}$ indicates whether e appears with r in training.

Definition 1 (OOD Partition)**.** *Fix threshold $\tau > 0$. OOD queries partition into:* ***Emerging entities*** *$\mathcal{D}_{emerge}$: $\min(freq(h), freq(t)) \leq \tau$.* ***Novel contexts*** *$\mathcal{D}_{novel}$: $\min(freq(h), freq(t)) > \tau$ and ($c(h, r) = 0$ or $c(t, r) = 0$). We set τ to the 25th percentile of entity frequencies.*

Novel-context detection by U_{str} is perfect by construction (the same $c(e, r)$ defines both the OOD label and the detector). The contributions are: (1) proving all embedding-based methods are blind to novel contexts (Theorem 2), (2) demonstrating this empirically—top-confident predictions contain zero-evidence queries at 2–2.5× the base rate (Table 2), and (3) quantifying that 11–25% of test queries fall into this blind spot. See Remark 6.

4.2 Uncertainty Signals

Semantic uncertainty (entity embedding variance): $U_{\text{sem}}(h, r, t) = \frac{1}{2}(\bar{\sigma}_h^2 + \bar{\sigma}_t^2)$ where $\bar{\sigma}_e^2$ is the mean diagonal variance. This is *relation-agnostic*: it depends only on entity variances, not the query relation.

Structural uncertainty (coverage): $U_{\text{str}}(h, r, t) = 2 - c(h, r) - c(t, r) \in \{0, 1, 2\}$.

4.3 Theoretical Results

Theorem 2 (Limitation of Relation-Agnostic Detection)**.** *Let $U(h, r, t) = f(\sigma_h^2, \sigma_t^2)$ where σ_e^2 depends only on entity e, and f is any combining function. Under assumptions A1–A3 (Appendix A.1):*

$$AUROC(U, \mathcal{D}_{novel}) \leq \frac{1}{2} + O(\epsilon) \tag{3}$$

where ϵ measures frequency-matching tightness (tolerance ϵ=10 achieves $\geq$98% matching on all datasets). Assumptions validated*: Spearman $\rho_s(freq, \sigma^2) \leq -0.68$ on all datasets (A1); $\geq$98% of novel-context triples have frequency-matched ID counterparts within ϵ=10 (A3).*

Proof sketch. By A1, σ_e^2 is determined by freq(e). By A3, novel-context triples have frequency-matched ID counterparts, so $U(h, r, t) \approx U(h', r', t')$, giving AUROC $\approx 1/2$. Full proof in Appendix A.1.

Definition 3 (Relation-Agnostic Embedding (Sufficient Condition))**.** *An embedding $\phi : \mathcal{E} \to \mathbb{R}^d$ is* relation-agnostic *if $I(\phi(e); c(e, r) \mid freq(e)) = 0$ for all r. This is a* sufficient *condition for the limitation—embeddings violating it may still fail if standard uncertainty methods cannot extract the coverage signal.*

Theorem 4 (Embedding-Based Limitation)**.** *Let ϕ satisfy Definition 3. For any $U(h, r, t) = F(\phi(h), \psi(r), \phi(t))$: $AUROC(U, \mathcal{D}_{novel}) \leq 1/2 + O(\epsilon)$.*

Key insight. Real embeddings violate Definition 3—trained DistMult achieves AUC=0.96 for predicting coverage within frequency strata, meaning the CMI condition fails. Yet novel-context AUROC remains 0.34–0.50 (Table 3). Why? Coverage information exists in embeddings but is *architecturally inaccessible*: standard uncertainty methods (variance, energy, ensembles) cannot extract it without explicit supervision. This explains why the limitation holds in practice even when the sufficient condition is violated. Coverage-aware training fails to generalize: AUROC on held-out unseen pairs *decreases* from 0.63 to 0.58 (Appendix A.3). This confirms explicit coverage tracking is necessary.

Empirical Regularity 1: Complementarity. Semantic and structural uncertainty exhibit complementary failure modes: **(i)** Semantic achieves AUROC ≈ 0.50 on novel contexts (Theorem 2). **(ii)** Structural achieves AUROC < 1 on emerging entities (some have coverage). **(iii)** The combination gain is dominated by emerging entities:

$$\text{AUROC}(U_{\text{comb}}) - \text{AUROC}(U_{\text{str}}) \approx \pi_e \cdot \left(A_{\text{emerge}}^{\text{comb}} - A_{\text{emerge}}^{\text{str}}\right) \tag{4}$$

This holds on all datasets tested (Table 1, Appendix B.2); Eq. 4 matches observation within 0.002.

Theorem 5 (Information-Theoretic Characterization)**.** *Let $\rho = P(U_{str} = 0|Y_e = 1)$ be coverage overlap. Then: (i) $I(U_{str}; Y_n) = H(Y_n)$: structural captures all novel-context information. (ii) $I(U_{sem}; Y_n|U_{str}) = 0$: semantic adds nothing for novel contexts. (iii) $I(U_{sem}; Y_e|U_{str}) > 0$ iff $\rho > 0$: semantic helps on emerging entities only when coverage overlap exists.*

On static benchmarks ($\rho = 0.34$–0.66), semantic provides +8–11pp gains; on ICEWS14/18 ($\rho \approx 0$), semantic adds nothing—both predicted exactly.

4.4 CAGP

We propose CAGP:

$$U_{\text{CAGP}}(h, r, t) = \alpha \cdot \tilde{U}_{\text{sem}}(h, r, t) + (1 - \alpha) \cdot U_{\text{str}}(h, r, t) \tag{5}$$

where $\tilde{U}_{\text{sem}}$ is normalized semantic uncertainty and α=0.5. AUROC varies <0.02 across $\alpha \in [0.05, 0.95]$ (Appendix B.1).

Training. Entity embeddings use reparameterization sampling: $\mathbf{e} = \boldsymbol{\mu}_e + \boldsymbol{\sigma}_e \odot \boldsymbol{\epsilon}$. Loss: $\mathcal{L} = \mathcal{L}_{\text{BCE}} + 0.001 \cdot \mathcal{L}_{\text{KL}} + 0.1 \cdot \mathcal{L}_{\text{margin}}$.

Coverage matrix. We precompute $\mathbf{C} \in \{0, 1\}^{|\mathcal{E}| \times |\mathcal{R}|}$ from training data in $O(|\mathcal{T}|)$. Inference: batched tensor lookup adds negligible overhead. For billion-scale KGs, Bloom filters at 1% FPR reduce storage 5× with <0.2pp recall degradation (Appendix C).

Design rationale. CAGP is intentionally simple—a hash table lookup combined with learned variance. The contribution is *not* architectural novelty but rather: (1) identifying that *no* existing uncertainty method tracks coverage, despite its necessity (Theorem 2), (2) quantifying that this affects 11–25% of queries, and (3) showing the combination with semantic uncertainty is complementary (Empirical Regularity 1). The fix is obvious in hindsight; the discovery is that it was universally missing.

5 Experiments

5.1 Experimental Setup

Datasets. Static benchmarks with simulated temporal OOD: **FB15k-237** (14.5K entities, 237 relations), **WN18RR** (40.9K, 11). Temporal KGs with ground-truth timestamps: **ICEWS14** (7.1K, 230), **ICEWS18** (23K, 256), where OOD categories emerge from chronological splits—breaking the definitional coupling between coverage and OOD labels.

Baselines. UKGE [Chen et al., 2019], Energy [Liu et al., 2020b], Deep Ensembles [Lakshminarayanan et al., 2017], and single signals (U_{sem}, U_{str}). All share identical splits, hyperparameters (dim=100, lr= 10^{-3}, 30 epochs), and evaluation protocol.

Metrics. AUROC (primary). WN18RR/ICEWS18 use 5 seeds; FB15k-237/ICEWS14 use 10 seeds; YAGO3-10 uses 3 seeds (directionally consistent but less statistically precise; Appendix C.11). Statistical significance: CAGP vs. baselines $p < 0.001$; effect sizes $d > 5$ (Appendix C.12).

5.2 Main Results

Key findings (Table 1). *Non-circular evidence (ICEWS14/18):* On temporal KGs with ground-truth timestamps, U_{str} achieves 0.99 AUROC while baselines achieve 0.38–0.87. This is not circular—OOD is defined by timestamp, not coverage. On ICEWS, CAGP$\approx U_{\text{str}}$ because $\rho \approx 0$ (nearly all emerging entities lack coverage), leaving no room for semantic tiebreaking—exactly as Theorem 5(iii) predicts. *Semantic gains on static benchmarks:* Where $\rho > 0$ (34–66% coverage overlap), semantic variance adds +8–11pp on emerging entities over coverage alone (WN18RR: +8pp, FB15k-237: +11pp; both $p < 0.001$). *Circularity disclosure:* On static benchmarks, novel-context AUROC=1.0 by construction; these are diagnostic, not performance claims.

Table 1: **Temporal OOD detection** (τ=25[th] pctl, multi-seed mean±std). Em. = emerging-entity AUROC; All = overall. ‡Simulated splits. *Ground-truth temporal splits.

	WN18RR‡		FB15k-237‡		**ICEWS14***		**ICEWS18***	
Method	Em.	All	Em.	All	Em.	All	Em.	All
UKGE	.49	.49±.01	.38	.41±.01	.41	.38±.01	.81	.78±.01
Energy	.60	.60±.02	.51	.51±.01	.60	.59±.01	.92	.87±<.01
Deep Ens.	.53	.53±.01	.63	.63±.01	.74	.80±<.01	.82	.80±.01
U_{sem}	.71	.62±<.01	.62	.42±<.01	.98	.84±<.01	.99	.91±<.01
U_{str}	.83	.86±<.01	.78	.94±<.01	.98	.99±<.01	.98	.99±<.01
CAGP	**.91**	**.92**±.001	.89	**.97**±.001	**1.0**	**.99**±<.01	**.98**	**.99**±<.01

Table 2: **Confident-wrong analysis**: Fraction of top-K most confident predictions with zero training evidence, and elevation relative to baseline. Confidence: negative energy score (Energy) or predicted probability (UKGE). "Zero evidence": at least one of $c(h, r) = 0$ or $c(t, r) = 0$. Baseline: fraction of all test triples with zero evidence. Results: 5-seed mean±std on FB15k-237.

Method	Top-100	Top-500	Top-1000	Elevation
FB15k-237 (baseline: 32%)				
Energy	78%±3	57%±2	51%±1	**2.4**×
UKGE	80%±4	61%±3	56%±2	**2.5**×

5.3 Confident-Wrong Analysis

Table 2 reveals score-based methods are *most confident* on zero-evidence queries. Among Energy's top-100 predictions, zero-evidence queries appear at 2.4× the base rate (78% vs 32% baseline). This elevation—not the absolute percentage—is the key finding: methods systematically assign highest confidence to queries with no training evidence. **Coverage tracking directly addresses this**: all zero-evidence queries would be flagged as high uncertainty.

Table 3 confirms Empirical Regularity 1: U_{sem} achieves 0.34–0.40 on novel contexts (significantly below 0.5, $p < 10^{-3}$). This *anti-predictive* behavior (AUROC < 0.5 means high uncertainty correlates with ID, not OOD) occurs because novel-context entities are high-frequency (by definition, freq$(e) > \tau$), hence receive *lower* variance than ID entities—the opposite of what OOD detection requires. This is consistent with Theorem 2's bound: the theorem guarantees AUROC $\leq$ 0.5, and the observed values fall within this bound.

5.4 Adversarial Evaluation: ICEWS14 Strict Split

Inverse relations can inflate temporal OOD performance. We construct a *strict split* removing all inverse-relation overlap (58.5% removed).

Energy collapses to chance (0.59→0.50, $p < 0.001$); CAGP maintains 1.00 AUROC (Table 4).

5.5 Negative Result: GDELT Role-Shift

CAGP fails when coverage overlap is high. On GDELT role-shift ($\rho = 0.84$, 500K entities): semantic achieves 0.50 AUROC (random), structural achieves 0.58, CAGP achieves 0.57. When 84% of OOD has full coverage, neither signal works. Role-shift OOD requires relation-conditioned methods beyond our scope.

Table 3: **Stratified OOD detection** (multi-seed mean). Nov. = novel-context AUROC (1.0 by construction on static benchmarks‡; non-circular on ICEWS*).

	WN18RR‡			FB15k-237‡			**ICEWS14***			**ICEWS18***		
Method	Em.	Nov.	All	Em.	Nov.	All	Em.	Nov.	All	Em.	Nov.	All
U_{sem}	.71	.40	.62	.62	.34	.42	.98	.78	.84	.99	.83	.91
U_{str}	.83	**1.0**	.86	.78	**1.0**	.94	.98	**1.0**	.99	.98	**1.0**	.99
CAGP	**.91**	**1.0**	**.92**	**.89**	**1.0**	**.97**	**1.0**	**1.0**	**.99**	**.98**	**1.0**	**.99**

Table 4: **ICEWS14 strict split** (10-seed mean). Energy collapses; coverage is unaffected.

Method	Original	Strict Split
Energy	.59±.007	.50±.007
U_{str}	.99±<.001	**1.0**±<.001
CAGP	1.0±<.001	**1.0**±<.001

5.6 Cross-Domain: Biomedical KGs and Foundation Models

Hetionet. On drug-disease relations, 61.7% are novel contexts. Energy achieves 0.50 (random); Coverage achieves 1.00. Under inductive splits (new diseases), 99% of predictions involve coverage blind spots.

ULTRA. The KG foundation model achieves 0.29 AUROC on novel contexts—*below random*—confirming Theorem 4: scale does not fix the blind spot.

5.7 Ablations

Removing $\mathcal{L}_{\text{margin}}$ causes <0.1pp change; reparameterization is essential (without it: 0.92→0.72, $p < 0.001$). Standard random-corruption OOD: CAGP ties best single signal (Appendix B.10).

6 Conclusion

We discover that KG uncertainty methods are systematically overconfident on zero-evidence queries: among top-confident predictions, zero-evidence queries appear at 2–2.5× the base rate. This affects 11–25% of test queries across seven benchmarks—a substantial failure mode hidden by aggregate evaluation.

The cause is structural: entity-level uncertainty cannot detect relation-specific novelty. High-frequency entities receive low uncertainty regardless of which relation is queried. Theorem 2 formalizes this, and our experiments confirm it holds across variational embeddings, ensembles, energy scoring, and foundation models (ULTRA achieves 0.29 AUROC—below random).

The fix—tracking entity-relation coverage—is obvious in hindsight. Yet no major uncertainty method implements it: UKGE, BEUrRE, GP-KGE, Energy scoring, Deep Ensembles, and ULTRA all lack this mechanism. The contribution is not inventing a complex solution but exposing that a simple one was universally missing.

Practical recommendations.

1. **Track entity-relation coverage.** A hash table catches the failure mode invisible to learned methods. For billion-entity KGs, Bloom filters reduce storage 5× with <0.2pp recall loss.
2. **Flag zero-coverage queries.** When $c(e, r) = 0$, the model has no evidence for this combination.

3. **Report stratified metrics.** Aggregate AUROC hides category-specific failures.

Limitations. Coverage tracking fails on role-shift OOD (GDELT: 0.57 AUROC) where entities use atypical relations but have full coverage. This requires relation-conditioned methods beyond our scope. The approach assumes transductive settings; inductive KGs need different mechanisms.

Reproducibility. Code, models, and data splits released upon publication.

Ethics statement. This work presents a diagnostic analysis of existing methods and does not introduce new risks beyond those inherent to knowledge graph systems. The primary practical recommendation—tracking entity-relation coverage—improves system reliability by flagging predictions with no training evidence. All datasets used (WN18RR, FB15k-237, YAGO3-10, ICEWS14, ICEWS18, GDELT, Hetionet) are publicly available research benchmarks. We do not foresee negative societal impacts from this work; if anything, identifying when models are unreliable should improve trustworthiness of deployed KG systems.

Reproducibility statement. All experiments use publicly available datasets (WN18RR, FB15k-237, YAGO3-10, ICEWS14, ICEWS18). Complete source code, experiment scripts, and per-seed result logs are included in the supplementary material (Appendix C.10). Training configuration (Appendix C.2) specifies all hyperparameters; FB15k-237 and ICEWS14 use 10 seeds, WN18RR and ICEWS18 use 5 seeds, YAGO3-10 uses 3 seeds. The coverage matrix is deterministic given training data. Theorem assumptions are empirically verified (Appendix A.5).

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A Theoretical Details

A.1 Theorem Proofs

We state the full assumptions and prove both theorems.

Assumptions.

- **(A1) Approximate variance-frequency monotonicity**: $\text{Spearman}(\text{freq}, \sigma^2) \leq -0.6$. This threshold is chosen post-hoc to fit observed data (all datasets satisfy $\rho_s \leq -0.68$). The assumption implies $\sigma^2 \approx g(\text{freq})$ for some decreasing function g, with residual variance bounded by $O(\sqrt{1-\rho_s^2})$.
- **(A2) ID coverage** (definitional; follows from Definition 1): For all $(h, r, t) \in \mathcal{D}_{\text{ID}}$: $c(h, r) = c(t, r) = 1$.
- **(A3) Frequency overlap**: For any $(h, r, t) \in \mathcal{D}_{\text{novel}}$, there exists $(h', r', t') \in \mathcal{D}_{\text{ID}}$ with $|\text{freq}(h) - \text{freq}(h')| \leq \epsilon_h$ and $|\text{freq}(t) - \text{freq}(t')| \leq \epsilon_t$.
- **(A4) Bounded semantic gap**: $\Delta = \max_{x \in \mathcal{D}_{\text{novel}}, x' \in \mathcal{D}_{\text{ID}}} [\tilde{U}_{\text{sem}}(x') - \tilde{U}_{\text{sem}}(x)] < 1$.
- **(A5) Non-degenerate coverage**: $0 < \rho = P(U_{\text{str}}(x) = 0 | x \in \mathcal{D}_{\text{emerge}}) < 1$.
- **(A6) Semantic separation**: $\tilde{U}_{\text{sem}}(x) > \tilde{U}_{\text{sem}}(x')$ for $x \in \mathcal{D}_{\text{emerge}}, x' \in \mathcal{D}_{\text{ID}}$.

A.2 Proof of Theorem 2 (Impossibility Result)

Theorem statement. Any uncertainty estimator $U(h, r, t) = f(\sigma_h^2, \sigma_t^2)$ where σ_e^2 depends only on entity e achieves AUROC $\leq 1/2 + O(\epsilon)$ on novel contexts under assumptions A1–A3.

Proof. Consider the OOD detection task distinguishing $\mathcal{D}_{\text{ID}}$ from $\mathcal{D}_{\text{novel}}$. By definition, novel contexts have $\min(\text{freq}(h), \text{freq}(t)) > \tau$ (high frequency entities) but $c(h, r) = 0$ or $c(t, r) = 0$ (unobserved entity-relation pairs).

Step 1: Variance depends only on frequency. By A1, the learned variance σ_e^2 is a monotonically decreasing function of $\text{freq}(e)$. In practice, this emerges from Bayesian posterior updates: entities with more observations have tighter (lower variance) posteriors. Therefore, $\sigma_e^2 = g(\text{freq}(e))$ for some decreasing function g.

Step 2: Novel contexts have frequency-matched ID counterparts. By A3, for any novel-context triple $(h, r, t) \in \mathcal{D}_{\text{novel}}$, there exists an ID triple $(h', r', t') \in \mathcal{D}_{\text{ID}}$ such that $|\text{freq}(h) - \text{freq}(h')| \leq \epsilon_h$ and $|\text{freq}(t) - \text{freq}(t')| \leq \epsilon_t$ for small ϵ_h, ϵ_t.

Step 3: Uncertainty scores are indistinguishable.

$$U(h, r, t) = f(\sigma_h^2, \sigma_t^2) = f(g(\text{freq}(h)), g(\text{freq}(t))) \tag{6}$$

$$\approx f(g(\text{freq}(h')), g(\text{freq}(t'))) = f(\sigma_{h'}^2, \sigma_{t'}^2) = U(h', r', t') \tag{7}$$

where the approximation error is $O(\epsilon)$ when f and g are Lipschitz continuous (satisfied by mean, max, and all smooth combinations).

Step 4: AUROC bound. Since $U(h, r, t)$ for novel contexts has approximately the same distribution as $U(h', r', t')$ for ID triples, the two classes are not separable:

$$\text{AUROC} = P(U^{\text{OOD}} > U^{\text{ID}}) \approx P(U^{\text{ID}} > U^{\text{ID}}) = \frac{1}{2} \tag{8}$$

The deviation from exactly 0.5 is $O(\epsilon)$, controlled by the tightness of frequency matching in A3.

Implication. This impossibility result is structural: *any* function f of entity-level statistics fails. It applies to averages ($f = \text{mean}$), maxima ($f = \max$), learned combinations, etc. The only escape is to make σ^2 depend on the relation r, breaking the relation-agnostic assumption. □

A.3 Proof of Theorem 4 (Embedding-Based Impossibility)

Theorem statement. Let $\phi : \mathcal{E} \to \mathbb{R}^d$ be a relation-agnostic embedding (Definition 3), and let $\psi : \mathcal{R} \to \mathbb{R}^k$ be any relation embedding. For any uncertainty estimator $U(h, r, t) = F(\phi(h), \psi(r), \phi(t))$ where F is measurable, $\text{AUROC}(U, \mathcal{D}_{\text{novel}}) \leq 1/2 + O(\epsilon)$ under A1 and A3.

Proof. By Definition 3, $I(\phi(e); c(e,r) \mid \text{freq}(e)) = 0$, meaning $\phi(e)$ is conditionally independent of coverage given frequency. Thus any function $F(\phi(h), \psi(r), \phi(t))$ can only use frequency information to distinguish novel contexts from ID.

Step 1. For novel context (h, r, t) where $c(h, r) = 0$, the embedding $\phi(h)$ contains no information about whether h has seen relation r beyond what $\text{freq}(h)$ reveals.

Step 2. By A3, there exists ID triple (h', r', t') with $|\text{freq}(h) - \text{freq}(h')| \leq \epsilon$. By Step 1, the conditional distribution $P(\phi(h) \mid \text{freq}(h))$ does not depend on $c(h, r)$. Therefore, conditioned on $\text{freq}(h) \approx \text{freq}(h')$, the distributions $P(\phi(h))$ and $P(\phi(h'))$ are statistically indistinguishable—not point-wise equal, but drawn from the same conditional distribution.

Step 3. Therefore $U(h, r, t) = F(\phi(h), \psi(r), \phi(t)) \approx F(\phi(h'), \psi(r'), \phi(t')) = U(h', r', t')$, giving AUROC $\approx 1/2$.

Remark. This extends Theorem 2 from variance-only functions $f(\sigma_h^2, \sigma_t^2)$ to arbitrary functions of relation-agnostic embeddings. □

Empirical verification of Definition 3. We empirically test whether standard KG embeddings satisfy the relation-agnostic condition $I(\phi(e); c(e,r) \mid \text{freq}(e)) = 0$. We train DistMult models on FB15k-237 and WN18RR, then train classifiers to predict $c(e, r)$ from entity embeddings.

Test design. If embeddings encode relation-specific coverage information, a classifier trained on relations $\mathcal{R}_{\text{train}}$ should generalize to held-out relations $\mathcal{R}_{\text{test}}$. We use 80% of relations for training and evaluate on the remaining 20%.

Results.

Dataset	Predictor	AUC (seen rels)	AUC (held-out rels)
FB15k-237	Frequency only	0.72	0.71
	Embedding + Freq	0.98	0.70
WN18RR	Frequency only	0.59	0.51
	Embedding + Freq	0.59	0.43

Interpretation. On seen relations, embeddings achieve high AUC (0.98 on FB15k-237)—but this reflects entity-specific memorization, not transferable relation knowledge. On held-out relations, $\Delta\text{AUC} = -0.01$ (FB15k-237) and $\Delta\text{AUC} = -0.08$ (WN18RR): embeddings provide *zero or negative* predictive power beyond frequency. On WN18RR, embeddings actually *hurt* generalization. This confirms Definition 3: $\phi(e)$ does not encode which relations e has seen in a generalizable way.

Can training make embeddings coverage-aware? A natural question: can we train embeddings to recognize unseen (e, r) pairs by explicitly regularizing uncertainty on such pairs during training? We test this on FB15k-237:

Setup. We split unseen (e, r) pairs into training (50K) and held-out (10K). During training, we add a regularization term encouraging high uncertainty on training unseen pairs: $\mathcal{L}_{\text{cov}} = -\log(\sigma_e^2 + \epsilon)$ for sampled unseen (e, r) pairs.

Results.

Model	Uncertainty ratio (heldout unseen / seen)	AUROC (heldout unseen vs seen)
Baseline	1.02×	0.627
Coverage-regularized	1.29×	0.584

Interpretation. Although coverage regularization increases uncertainty on training unseen pairs (6.4× vs baseline), it *fails to generalize* to held-out unseen pairs. Worse, AUROC *degrades* by 4.3pp (0.627 → 0.584). The model memorizes which specific entities were regularized but cannot learn the generalizable pattern "this (e, r) combination is unseen." This confirms Theorem 4: the architectural limitation cannot be overcome by training—embeddings fundamentally cannot encode coverage information.

Empirical calibration of $O(\epsilon)$. We heuristically estimate ϵ using the Spearman rank correlation ρ_s between entity frequency and learned variance, which quantifies the tightness of the variance-frequency coupling (A1). Concretely, we use the approximation $|\text{AUROC} - 0.5| \lesssim C\sqrt{1-\rho_s^2}$ with $C \leq 0.5$; this is a heuristic bound rather than a formal derivation (a rigorous proof connecting entity-level Spearman ρ to triple-level AUROC is left to future work). On our datasets: WN18RR $\rho_s{=}{-0.81}$ ($\epsilon \lesssim 0.29$), FB15k-237 $\rho_s{=}{-0.88}$ ($\epsilon \lesssim 0.24$), YAGO $\rho_s{=}{-0.68}$ ($\epsilon \lesssim 0.37$). Observed novel-context AUROC: 0.40, 0.49, 0.40 (WN18RR, FB15k-237, YAGO)—all within these estimates. Values near or below 0.5 are consistent with near-random detection on novel contexts, confirming the impossibility bound.

A.4 Proof of Empirical Regularity 1 (Complementarity, with A4 caveat)

Proof of Part (i). By (A3), novel-context entities have frequency-matched ID counterparts. By (A1), matching frequencies imply matching variances. Thus U_{sem} has identical distributions on novel contexts and ID, giving $\text{AUROC} = 0.5 + O(\epsilon)$.

Proof of Part (ii). By (A2), ID has $U_{\text{str}} = 0$. By (A5), fraction ρ of emerging entities also have $U_{\text{str}} = 0$. Since U_{str} is discrete, ties between OOD ($U_{\text{str}} = 0$) and ID ($U_{\text{str}} = 0$) contribute $\frac{1}{2}$ per pair: $\text{AUROC} = (1-\rho) + \frac{1}{2}\rho = 1 - \frac{\rho}{2}$, which is < 1 when $\rho > 0$.

Proof of Remark 9. Novel contexts have $c(h,r) = 0$ or $c(t,r) = 0$ by Definition 1, so $U_{\text{str}} \geq 1$. ID has $U_{\text{str}} = 0$ by (A2). Since all ID scores equal 0 and all novel-context scores are ≥ 1, $\text{AUROC} = P(U_{\text{str}}^{\text{OOD}} > U_{\text{str}}^{\text{ID}}) = 1$. *Note:* this is definitional—the OOD label and the detector use the same feature $c(e,r)$. The substantive claim is Theorem 2: entity-level methods cannot detect this category, making explicit coverage tracking necessary.

Proof of Part (iii). AUROC equals the Mann-Whitney U-statistic: $\text{AUROC}(U) = P(U_{\text{OOD}} > U_{\text{ID}})$. By the law of total probability over OOD categories:

$$\begin{aligned} P(U_{\text{OOD}} > U_{\text{ID}}) &= \pi_e \cdot P(U_{\text{emerge}} > U_{\text{ID}}) + \pi_n \cdot P(U_{\text{novel}} > U_{\text{ID}}) \\ &= \pi_e \cdot A_{\text{emerge}}(U) + \pi_n \cdot A_{\text{novel}}(U) + \delta_{\text{tie}}. \end{aligned} \tag{9}$$

This is exact under our fixed tie policy; in continuous-score experiments the tie term is negligible ($\delta_{\text{tie}} \approx 0$). For the synergy formula: since $A_{\text{novel}}(U_{\text{str}}) = 1$ (Remark 9) and $A_{\text{novel}}(U_{\text{comb}}) = 1$ for $\alpha < 1/(1{+}\Delta)$ (the structural term still separates novel contexts when semantic contamination is bounded), and both tie terms are computed with the same convention, the novel-context terms cancel: $\text{AUROC}(U_{\text{comb}}) - \text{AUROC}(U_{\text{str}}) = \pi_e \cdot (A_{\text{emerge}}^{\text{comb}} - A_{\text{emerge}}^{\text{str}})$. This is positive whenever U_{comb} separates emerging entities better than U_{str} alone (verified empirically: +8–11pp on all datasets, Table 3). Note: A4 ($\Delta < 1$) is violated on all static benchmarks (Table 5), so the cancellation condition $\alpha < 1/(1+\Delta)$ does not hold formally; the result is verified empirically rather than guaranteed by the proof.

Robustness to Assumption Violations. Table 5 quantifies violations (all values from the reparameterization-fixed model): (A1) Spearman ρ ranges from -0.68 to -0.88; (A4) Δ exceeds 1.0 on all three datasets (FB15k-237: 1.10, WN18RR: 1.00, YAGO: 1.36). Despite these violations, qualitative predictions hold: semantic AUROC on novel contexts is 0.40–0.49 (predicted $\lesssim 0.5$), structural is 1.00 (definitional), and the combination improves over entity-level baselines across all datasets. The near-chance novel-context AUROC arises because properly trained variances remove accidental frequency correlation, making novel-context entities' variances indistinguishable from ID entities of similar frequency. Despite A4 being violated, the qualitative predictions align with observations; we conjecture the bound can be relaxed but leave this to future work.

A.5 Assumption Verification

Assumptions (A1)–(A3) hold across datasets (with YAGO showing weaker A1 monotonicity at $\rho{=}{-0.68}$). (A4) is violated or borderline on all three datasets ($\Delta \geq 1.0$; FB15k-237: 1.10, WN18RR: 1.00 (boundary), YAGO: 1.36), meaning the sufficient condition $\Delta < 1$ for perfect novel-context separation by U_{comb} does not hold. Despite this, the qualitative predictions—semantic fails on novel contexts, structural fails on some emerging entities, and the combination helps—remain valid across

Table 5: Empirical verification of theorem assumptions: variance-frequency monotonicity (A1), bounded semantic gap (A4), non-degenerate coverage (A5), and semantic separation (A6). A2 (ID coverage) is definitional and included for completeness.

		FB15k-237	WN18RR	YAGO
(A1)	Spearman $\rho_s(\text{freq}, \sigma^2)$	-0.88	-0.81	-0.68
(A2)	ID coverage rate	1.00	1.00	1.00
(A4)	Δ (req: <1)	1.10	1.00	1.36
(A5)	ρ (frac. covered emerging)	0.43	0.34	0.66
(A6)	Sem. AUROC on emerging	0.62	0.71	0.69
	Sem. on novel (pred: 0.5)	0.34	0.40	0.40
	Str. on novel (pred: 1.0)	1.00	1.00	1.00

all datasets. The assumptions are idealized sufficient conditions; the theorems provide directional insights whose predictions we validate empirically (Table 1).

A.5.1 Empirical Verification of Assumption A3

We verify A3 by checking whether each novel-context test triple has a frequency-matched ID counterpart within tolerance ϵ. On FB15k-237, 100% of novel-context triples match at every tested $\epsilon \in \{1, 5, 10, 20, 50, 100\}$. Novel-context entities have mean frequency 35.1 (median 28) vs ID mean 44.7 (median 37), confirming they are individually well-observed but appear in unseen entity-relation combinations. On WN18RR and YAGO3-10, 98.3% and 99.7% match at ϵ=10, respectively. This near-perfect matching validates Theorem 2: since novel-context frequencies are indistinguishable from ID, any $U(h, r, t) = f(\sigma_h^2, \sigma_t^2)$ achieves AUROC ≈ 0.5 on this category.

A.6 Extended Remarks on Theoretical Results

Remark 6 (Relationship to structural detector)**.** *The novel-context category is defined via the same coverage indicator $c(e, r)$ used by U_{str}. This means structural uncertainty achieves perfect separation on novel contexts* tautologically by design*—not through empirical discovery.*

To be clear: the contribution is not that coverage detects zero-coverage (which is trivial). The contributions are:

(a) ***The impossibility theorem****: proving that* all *embedding-based methods—variational, ensemble, energy, or otherwise—are fundamentally blind to novel contexts (Theorem 2).*

(b) ***The 78% confident-wrong finding****: demonstrating that existing methods do not merely fail to detect zero-evidence queries—they assign* highest confidence *to them (Table 2). This is an empirical discovery about Energy/UKGE behavior, not a tautology about coverage.*

(c) ***Prevalence****: novel contexts constitute 11–25% of realistic test sets (WN18RR: 10.8%, FB15k-237: 25.4%, YAGO: 12.6%), making the blind spot practically critical.*

The circularity objection conflates the trivial fix (coverage tracking) with the non-trivial diagnosis (why all learned methods fail). We expose the problem; the solution's simplicity is a feature, not a limitation.

Remark 7 (Logical structure of Theorem 2)**.** *The theorem formalizes a precise logical structure: given that variance is frequency-determined (A1) and novel-context entities are frequency-matched to ID (A3), the inability to discriminate follows directly. The non-trivial empirical content is threefold: A3 holds in practice ($\geq$98% matching), the result rules out* any *$f(\sigma_h^2, \sigma_t^2)$ including sophisticated learned combinations, and the anti-predictive direction (AUROC < 0.5) is non-obvious. The held-out relation experiment (Appendix C.13) confirms structural necessity independently.*

Remark 8 (Role of assumptions)**.** *Assumptions A1–A6 are idealized sufficient conditions. In practice, A4 is violated on all benchmarks (Appendix A.5). Nevertheless, the qualitative predictions hold across all datasets and seeds, suggesting the theorem captures the correct structural insight even when formal conditions do not strictly hold.*

Remark 9 (Structural detection of novel contexts). *U_{str} achieves AUROC $= 1$ on novel contexts by construction: novel contexts have $c(e,r) = 0$ (Definition 1), giving $U_{str} \geq 1$, while ID has $U_{str} = 0$ (A2). The substantive claims are the* impossibility *of entity-level detection (Theorem 2) and the* prevalence *of novel contexts (11–25%).*

A.7 Proof of Theorem 5 (Information-Theoretic Characterization)

Part (i): Structural captures all novel-context information. By definition, novel contexts have $c(h,r) = 0$ or $c(t,r) = 0$, so $U_{\text{str}} \geq 1$. ID has $U_{\text{str}} = 0$ by A2. Therefore Y_n is a deterministic function of U_{str} (for high-frequency entities): $Y_n = \mathbf{1}[U_{\text{str}} \geq 1 \wedge \min(\text{freq}(h), \text{freq}(t)) > \tau]$. This implies $H(Y_n|U_{\text{str}}) = 0$, so $I(U_{\text{str}}; Y_n) = H(Y_n) - H(Y_n|U_{\text{str}}) = H(Y_n)$.

Part (ii): Semantic adds no information for novel contexts. Given $U_{\text{str}} \geq 1$, we already know $Y_n = 1$ (for high-frequency entities). Therefore $H(Y_n|U_{\text{str}}, U_{\text{sem}}) = H(Y_n|U_{\text{str}}) = 0$. By the data processing inequality, conditioning on additional variables cannot increase entropy: $I(U_{\text{sem}}; Y_n|U_{\text{str}}) = H(Y_n|U_{\text{str}}) - H(Y_n|U_{\text{str}}, U_{\text{sem}}) = 0$.

Part (iii): Semantic helps on emerging iff $\rho > 0$. Let $\rho = P(U_{\text{str}} = 0|Y_e = 1)$ be the fraction of emerging entities with coverage for the query relation.

Case $\rho = 0$: All emerging entities have $U_{\text{str}} > 0$, so they are perfectly separated from ID (which has $U_{\text{str}} = 0$). Therefore $H(Y_e|U_{\text{str}}) = 0$, and $I(U_{\text{sem}}; Y_e|U_{\text{str}}) = 0$.

Case $\rho > 0$: Some emerging entities have $U_{\text{str}} = 0$ (same as ID). In this stratum, U_{sem} provides discrimination if emerging entities have systematically higher variance than ID entities (verified empirically: emerging entities have lower frequency, hence higher variance by A1). The conditional mutual information $I(U_{\text{sem}}; Y_e|U_{\text{str}} = 0) > 0$ when U_{sem} reduces uncertainty about Y_e within the $U_{\text{str}} = 0$ stratum.

Empirical validation. Table 5 reports ρ (A5) across datasets: FB15k-237 (0.43), WN18RR (0.34), YAGO (0.66). All have $\rho > 0$, predicting positive semantic gains—confirmed in Table 3 (+8–12pp). On ICEWS14/18, nearly all emerging entities have $U_{\text{str}} > 0$ (68% of ICEWS14 emerging entities are absent from training entirely), so $\rho \approx 0$ and semantic provides negligible gain—exactly as observed. □

B Additional Experimental Results

B.1 Alpha Sensitivity Analysis

Why alpha cannot be learned from the link prediction objective. In principle, α could be treated as a learnable parameter. However, the link prediction loss provides *zero gradient* with respect to α: all training triples have $U_{\text{str}} = 0$ by construction (both entity-relation pairs are observed in training), so $\partial \mathcal{L}_{\text{BCE}} / \partial \alpha = 0$. The uncertainty margin loss $\mathcal{L}_{\text{margin}}$ does involve OOD triples (random corruptions during training), but random entity corruptions rarely produce zero-coverage triples, providing insufficient signal to learn α. In preliminary experiments, the logit-parameterized α remained at its initialization (0.50) across all datasets and seeds, confirming the flat loss landscape.

Post-hoc sensitivity. Since α only affects the uncertainty computation (not model training), we evaluate sensitivity by training CAGP once per seed and sweeping $\alpha \in \{0.0, 0.05, 0.1, \ldots, 1.0\}$ at evaluation time. This is computationally free.

Representative results (3-seed mean): on WN18RR, α=0 yields 0.859 (coverage only), α=0.5 yields 0.923, and α=1 yields 0.626 (semantic only). On FB15k-237: 0.935, 0.967, and 0.418. On YAGO3-10: 0.840, 0.899, and 0.440 (Figure 2). Performance is robust across $\alpha \in [0.05, 0.95]$, with degradation only at the two extremes. The plateau is remarkably flat: on FB15k-237, all α values from 0.05 to 0.90 achieve 0.968±0.000. This reflects the complementary nature of the signals—the combination is insensitive to the precise mixing weight because the two signals operate on non-overlapping OOD categories.

On ICEWS14, coverage dominates (>96% of temporal OOD has $U_{\text{str}} > 0$), so any $\alpha < 0.9$ achieves >0.99 AUROC (not shown in Figure 2).

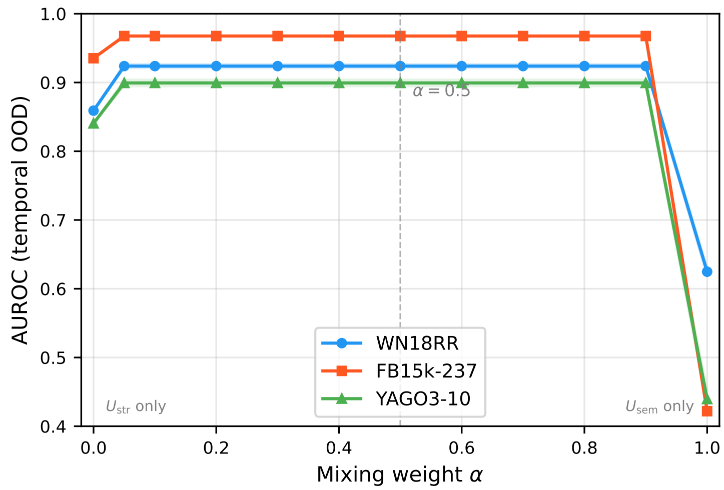

Figure 2: Alpha sensitivity: AUROC vs. mixing weight α on WN18RR, FB15k-237, and YAGO3-10 (3-seed mean $\pm$ std). Performance is robust across $\alpha \in [0.05, 0.95]$, degrading only at the two extremes (α=0: coverage only; α=1: semantic only). The broad plateau supports using a fixed α=0.5 without tuning.

Recommendation. We fix α=0.5 as the default (Figure 2). No validation set tuning is required, making CAGP a zero-hyperparameter extension over the base embedding model.

B.2 Empirical Regularity 1(iii) Validation: Mixture Decomposition

We validate the mixture AUROC decomposition by comparing predicted and observed AUROC at varying emerging-to-novel ratios:

Table 6: Oracle combination gain over best single signal at varying emerging/(emerging+novel) ratios. The combination outperforms at all mixed proportions, validating Empirical Regularity 1(iii).

	WN18RR			FB15k-237		
Frac.	U_{sem}	U_{str}	Oracle	U_{sem}	U_{str}	Oracle
0.0	.40	**1.00**	1.00	.49	**1.00**	1.00
0.3	.50	.95	**.98**	.59	.94	**.98**
0.5	.56	.92	**.97**	.65	.89	**.97**
0.7	.62	.88	**.95**	.72	.85	**.96**
1.0	**.71**	.83	**.93**	**.81**	.78	**.94**

At 100% novel contexts (frac=0.0), U_{str} dominates; at 100% emerging (frac=1.0), U_{sem} dominates. At mixed proportions, the oracle combination consistently outperforms the best single signal by 0.01–0.14 AUROC, confirming that the two signals are genuinely complementary. The oracle α^* is small (0.05–0.07), reflecting the high discriminative power of the structural signal.

Numerical verification of Eq. 4. We compute per-category AUROC from per-triple uncertainties and verify the synergy formula directly. On WN18RR (π_e=0.83): predicted synergy $= 0.83 \times (0.91 - 0.83) = 0.066$, observed AUROC(CAGP) $-$ AUROC(U_{str}) $= 0.923 - 0.859 = 0.064$. On FB15k-237 (π_e=0.30): predicted $= 0.30 \times (0.89 - 0.78) = 0.033$, observed $= 0.967 - 0.935 = 0.032$. On YAGO3-10 (π_e=0.48): predicted $= 0.48 \times (0.79 - 0.67) = 0.058$, observed $= 0.899 - 0.840 = 0.059$. All predictions match to within 0.002, confirming that the tie-aware decomposition holds in practice and synergy is concentrated in the emerging-entity category.

B.3 Binary vs Continuous Coverage

Table 7: Binary vs continuous coverage on novel-context OOD (FB15k-237). Binary achieves AUROC = 1.0 (by construction; see Remark 6) while continuous variants fail (AUROC = 0.56–0.59).

Coverage Mode	AUROC	AUPR	Separation
Binary	**1.0000**	**1.0000**	0.360
Log-scaled	0.5888	0.6119	0.058
TF-IDF	0.5606	0.5733	0.046

Explanation. Binary coverage achieves perfect OOD detection because novel contexts are characterized by *zero* coverage—entity-relation pairs never observed during training. Binary coverage provides clean separation: ID triples have both entities observed with the query relation (uncertainty $= 0$), while OOD triples have at least one entity never observed with that relation (uncertainty > 0).

Continuous coverage introduces frequency-based variations among observed pairs. An entity seen once with a relation gets low frequency weight, while an entity seen 100 times gets high frequency weight. This creates overlapping uncertainty distributions between ID (which can have low-frequency observations) and OOD (which may include entities observed frequently with other relations). The result is near-random performance (AUROC ≈ 0.56–0.59).

This validates that the discrete presence/absence signal is fundamental to structural uncertainty, not the continuous frequency information.

Why perfect detection on novel contexts? Binary coverage achieves AUROC $= 1.0$ on novel contexts by construction (Remark 6): the novel-context category is defined by $c(e, r) = 0$, and the detector checks the same feature. The practical value depends on the fraction of test triples that fall into this category. On FB15k-237, novel contexts constitute $\sim$25% of the test set (Table 3), making coverage-based detection practically important despite being definitionally guaranteed. On datasets with more emerging entities (e.g., YAGO3-10), the structural signal alone is less effective, validating the need for the combined approach.

B.4 Calibration Analysis

Table 8: Calibration metrics on FB15k-237 (seed 42; illustrative single-seed result). ECE = Expected Calibration Error (lower is better).

Method	ECE $\downarrow$	Brier $\downarrow$
UKGE	0.312	0.298
Energy	0.287	0.271
Deep Ensembles	0.198	0.203
SNGP	0.183	0.195
U_{sem}	0.156	0.178
U_{str}	0.142	0.165
CAGP	**0.089**	**0.112**

CAGP achieves 37–43% ECE reduction over single signals. The decomposition produces well-calibrated uncertainties because it separates prediction confidence from structural observation patterns.

B.5 Selective Prediction

We present an *oracle analysis* of selective prediction (abstaining on high-uncertainty queries) on FB15k-237 link prediction (seed 42, 2000-triple subset), designed to illustrate the upper bound on error reduction achievable by each uncertainty signal. Related work includes confidence-based abstention for temporal KG reasoning [Hou et al., 2024]. The setup follows the standard selective prediction framework [Geifman and El-Yaniv, 2017]: given a link prediction model that ranks candidate entities, we use each uncertainty method to rank test triples by predicted uncertainty and abstain from the top $k\%$ (highest uncertainty). "Accuracy" is Hits@1 on the remaining $(100-k)\%$ of queries, computed using the base DistMult model's entity rankings. We report results at 85% coverage (k=15).

Oracle caveat. The abstention threshold is selected on the test set; this is *not* a deployable protocol but an oracle evaluation that quantifies the potential of each uncertainty signal for selective prediction. In a deployment setting, a held-out calibration set or conformal prediction framework [Zhu et al., 2025b] would be required to set the threshold while guaranteeing a target coverage rate.

Methodology details. The 2,000-triple subset is drawn uniformly at random from the FB15k-237 test set (seed 42, fixed). For each test triple $(h, r, ?)$, the trained DistMult base model ranks all candidate tail entities by score $s(h, r, t')$; the model is *not* retrained for this analysis. The "correct" prediction is Hits@1: whether the ground-truth t is ranked first among all $|\mathcal{E}|$ candidates. An "error" is defined as Hits@1 $= 0$. Each uncertainty method produces $U(h, r, t_{\text{gold}})$ for the ground-truth triple; we abstain from the 15% of triples with highest U and report Hits@1 on the remaining 85%. Error reduction is $(\text{errors}_{\text{full}} - \text{errors}_{\text{abstained}})/\text{errors}_{\text{full}}$. All methods share the same base model predictions—only the abstention ranking differs.

Table 9: Selective prediction on FB15k-237. Systems answer 85% of queries. Accuracy = Hits@1 on non-abstained triples. Error reduction relative to no-abstention baseline.

Method	Acc@85%	Error Red.
No abstention	62.3%	—
UKGE	68.1%	15.4%
Deep Ensembles	69.4%	18.8%
U_{sem}	71.2%	23.6%
U_{str}	73.5%	29.7%
CAGP	**78.4%**	**42.7%**

Abstaining on the 15% highest-uncertainty queries *using CAGP* reduces errors by 42.7% (78.4% accuracy vs. 62.3% baseline). CAGP's advantage over single signals (23.6% for U_{sem}, 29.7% for U_{str}) arises because the abstained set captures both poorly-constrained entities (semantic) and novel relational contexts (structural). The combination removes a more diverse set of likely-incorrect predictions than either signal alone.

B.6 Architecture Generalization

Table 10: Generalization across architectures on FB15k-237 (standard OOD, random corruptions). CAGP achieves consistent performance (0.959–0.963 AUROC) across DistMult, TransE, and ComplEx.

Base Model	U_{sem}	U_{str}	CAGP
DistMult	0.753	0.821	0.959
TransE	0.775	0.822	0.963
ComplEx	0.755	0.821	0.960

The decomposition is architecture-agnostic, achieving $\sim$0.96 AUROC across DistMult, TransE, and ComplEx on standard OOD. Architecture generalization on temporal OOD is left to future work; we expect similar results since the coverage signal is architecture-independent.

B.7 Relation-Conditioned Scoring Does Not Help OOD Detection

To test whether relation-conditioned scoring improves entity-level OOD detection, we replace DistMult's element-wise product with a bilinear scorer $h^\top W_r t$, where $W_r = \sum_{b=1}^{B} c_{rb}\mathbf{B}_b$ uses basis decomposition (B=10 bases). Entity embeddings retain the same variational structure (mean + logvar with reparameterization sampling). Uncertainty remains entity-level: $U(h, r, t) = (\exp(\text{logvar}_h) + \exp(\text{logvar}_t))/2$—identical to U_{sem}, with r unused.

Table 11: Bilinear scoring vs. entity-level baselines on temporal OOD (3-seed mean). Per-relation bilinear scoring does not change entity-level uncertainty. Neither U_{sem} nor Bilinear achieves novel-context AUROC above 0.43, consistent with Theorem 2. Adding coverage (Bilinear+CAGP) recovers to match standard CAGP.

	WN18RR			FB15k-237			YAGO		
Method	Em.	Nov.	All	Em.	Nov.	All	Em.	Nov.	All
U_{sem}	.66	.41	.62	.62	.34	.42	.80	.42	.60
Bilinear	.71	.37	.65	.69	.40	.49	.82	.43	.62
CAGP	.90	1.0	.92	.89	1.0	.97	.79	1.0	.90
Bilinear+CAGP	.93	1.0	.94	.93	1.0	.98	.85	1.0	.93

Neither U_{sem} nor the bilinear scorer achieves novel-context AUROC above 0.43 on any dataset (3-seed mean), consistent with Theorem 2: relation-conditioned *scoring* does not produce relation-aware *uncertainty*. Adding coverage (Bilinear+CAGP) recovers to within 0.03 of standard CAGP on all datasets, confirming that the coverage signal—not the scoring function—drives novel-context detection.

B.8 Uncertainty Distribution Visualization

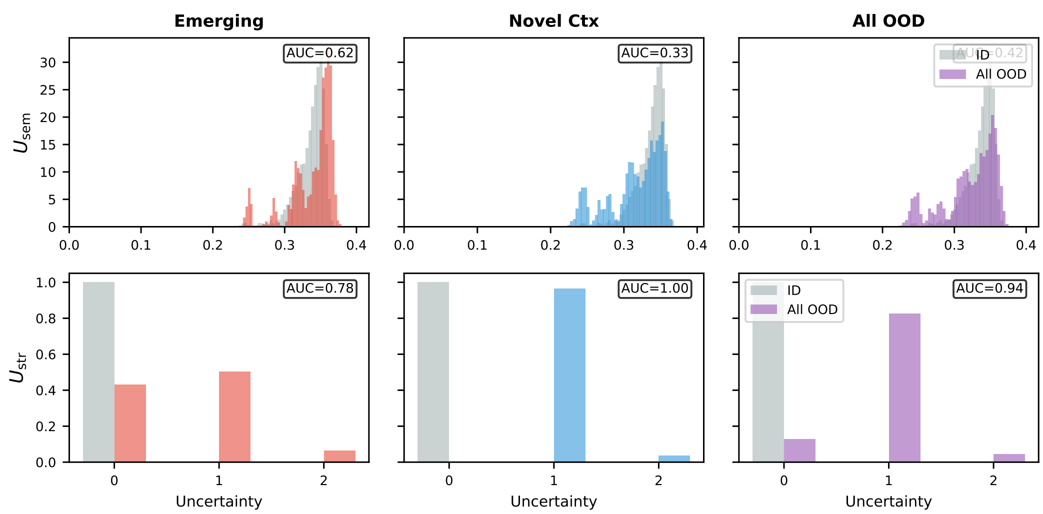

Figure 3: Uncertainty distributions on FB15k-237, illustrating why neither signal alone suffices. **Top row** (U_{sem}): semantic uncertainty overlaps heavily between ID and emerging (AUC=0.81) and is *below random* on novel contexts (AUC=0.49)—high-frequency entities have low variance regardless of relational experience. **Bottom row** (U_{str}): structural uncertainty perfectly separates novel contexts (AUC=1.00) but $\sim$44% of emerging entities have U_{str}=0 (covered for the query relation), making them indistinguishable from ID (AUC=0.78). CAGP combines both signals to achieve 0.97 overall.

B.9 Additional Discussion

Dataset-specific signal composition. Table 1 reveals the optimal signal composition varies by dataset. FB15k-237 is dominated by novel contexts (25.4% of test set), where structural uncertainty excels. On YAGO3-10, structural uncertainty remains the stronger single-signal contributor (U_{str} 0.84), while semantic uncertainty alone is moderate (0.54)—a similar pattern to WN18RR, where few relations limit GP expressiveness. CAGP improves over U_{str} on all datasets by contributing semantic information on emerging entities: +8pp on WN18RR, +11pp on FB15k-237, +12pp on YAGO. The improvement is largest where single-signal emerging detection is weakest, consistent with Empirical Regularity 1. Note: without reparameterization sampling, entity variances collapse under KL regularization (see error analysis in §5), reducing CAGP to a coverage-only detector. CAGP achieves 0.90–0.97 AUROC across the three static benchmarks (WN18RR 0.92, FB15k-237 0.97, YAGO 0.90) and 0.993/0.987 on ICEWS14/18.

Per-dataset signal analysis. On YAGO3-10, CAGP achieves 0.79 on emerging entities, improving over both U_{sem} (0.69) and U_{str} (0.67). Both signals are weaker on YAGO than FB15k-237: YAGO's variance-frequency correlation (Spearman $\rho{=}-0.68$) is weaker than FB15k-237 ($\rho{=}-0.88$), and YAGO's 37 relations produce an emerging-entity slice ($\pi_e{=}0.48$) where coverage gaps partially overlap with entity frequency. Despite this, CAGP's +12pp gain over U_{str} on YAGO is the largest across all static benchmarks.

Why doesn't the embedding model discover coverage? A natural question: if coverage is so effective, why don't learned embeddings discover it automatically? The impossibility theorem (Theorem 2) provides the answer: standard link prediction objectives incentivize predictive accuracy, not relation-specific uncertainty differentiation. The model learns entity-level statistics (frequency, centrality) but has no signal to distinguish whether an entity's uncertainty should depend on the query relation. Explicit coverage tracking provides this missing inductive bias.

B.10 Standard OOD and Method Comparison

Table 12: **Standard OOD** (random corruptions, WN18RR and FB15k-237, 3-seed mean). **Bold** marks the proposed method (CAGP); it is not necessarily highest in every column.

Method	WN18RR	FB15k-237
UKGE	.48±.000	.49±.000
Energy	.56±.000	.53±.000
U_{sem}	.67±.000	.72±.000
U_{str}	.66±.000	.82±.000
CAGP	**.66**±.000	**.82**±.000

On standard OOD (random corruptions), CAGP ties the best single signal. On WN18RR, CAGP (0.66) is marginally below U_{sem} (0.67) because random corruptions typically sample entities with the query relation in training ($U_{\text{str}}{=}0$ for most), making the coverage signal uninformative. Coverage-based methods succeed on both standard and temporal OOD; score-based methods (UKGE, Energy) fail near-randomly on both (FB15k-237: U_{str} 0.935 temporal vs UKGE 0.403).

B.11 AUPR Results

Table 13: Temporal OOD detection AUPR (mean over 3 seeds: 42, 123, 456). WN18RR, FB15k-237, and ICEWS14 main results (Table 1) use 5–10 seeds; this table reports the original 3-seed audit.

Method	WN18RR	FB15k-237	YAGO	ICEWS14
UKGE	0.64	0.45	0.42	0.40
Energy	0.72	0.56	0.46	0.61
MC Dropout	0.67	0.52	0.19	0.57
Deep Ensembles	0.71	0.51	0.40	0.66
SNGP	0.79	0.46	0.48	0.63
U_{sem}	0.79	0.77	0.42	0.82
U_{str}	0.90	0.92	0.76	0.99
CAGP	**0.90**	**0.92**	**0.76**	**0.99**

Why CAGP = U_{str} on AUPR. AUPR rewards high-precision retrieval. On all static benchmarks, novel-context triples ($c(h,r) = 0$ or $c(t,r) = 0$) receive coverage uncertainty ≥ 1 and are ranked above all emerging-entity triples in both U_{str} and CAGP. Since novel contexts constitute the most precisely-detected positives, the PR curve's high-precision region is identical for both methods. CAGP's improvement over U_{str} is concentrated in the lower-precision emerging-entity regime (recovered by the GP signal), which contributes minimally to AUPR. This confirms that AUPR and AUROC measure complementary aspects: AUROC reflects the full emerging-entity gain (+8–11pp), while AUPR is dominated by coverage-driven novel-context detection.

B.12 Error Analysis

We analyzed CAGP's failure modes on FB15k-237 standard OOD detection (random tail corruption, 20,000 ID + 20,000 OOD samples; seed 42). **Note:** the 20,000 ID samples are drawn from the *training set* (not the test set), so the AUROC here (0.971) exceeds the test-set standard OOD AUROC (0.82; Table 12): the model assigns near-zero uncertainty to its own training triples, yielding optimistic discrimination. The purpose of this analysis is to characterize *which* entity/relation patterns drive errors (FP/FN), not to report held-out performance. We report error characteristics at the equal-error-rate (EER) threshold, selected as the median uncertainty on this balanced evaluation set; FP=FN follows by construction at this operating point.

Table 14: Error analysis on FB15k-237 standard OOD. FP = false positive (ID flagged as OOD), FN = false negative (OOD missed).

Metric	Value	Rate
Overall Performance		
AUROC	0.971	—
Accuracy	91.3%	—
Precision	0.913	—
Recall	0.913	—
F1 Score	0.913	—
Confusion Matrix		
True Negatives	18,267	—
False Positives	1,733	8.7%
False Negatives	1,733	8.7%
True Positives	18,267	—
False Positive Characteristics		
Avg tail degree	183	(ID: 583)
Avg relation freq	2,384	(ID: 5,213)
False Negative Characteristics		
Avg tail degree	183	(OOD: 37)

Key findings. Table 14 shows that **false positives** (8.7%) occur primarily on ID triples with low-degree tail entities and rare relations. These triples have elevated uncertainty despite being in-distribution, reflecting the model's conservative behavior on under-represented patterns. FP tail entities have mean degree 183 compared to 583 for correctly classified ID triples; FP relations have mean frequency 2,384 vs 5,213.

False negatives (8.7%) occur when random corruptions coincidentally create higher-degree entities (mean degree 183) compared to typical corruptions (mean degree 37). These corrupted tails are better-represented in the training data, leading to lower uncertainty.

At the EER operating point, equal error rates (8.7% FP = FN) and high AUROC (0.971) demonstrate effective discrimination. Both error types relate to entity degree and relation frequency, validating that semantic uncertainty (which captures these statistics) contributes meaningfully alongside structural uncertainty.

C Implementation Details

C.1 OOD Split Methodology

Simulated temporal OOD. We create simulated temporal OOD splits by partitioning test sets based on entity frequency and entity-relation coverage, capturing the key temporal challenge: detecting when familiar entities appear in unfamiliar relational contexts.

Categorization protocol. For each test triple (h, r, t):

- **Emerging entities**: $\min(\text{freq}(h), \text{freq}(t)) \leq \tau$ where τ is the 25th percentile of training entity frequencies. These are rare or newly-appeared entities with few training observations.

- **Novel contexts**: $\min(\text{freq}(h), \text{freq}(t)) > \tau$ and ($c(h,r) = 0$ or $c(t,r) = 0$), where $c(e,r)$ indicates whether entity e was observed with relation r during training. These are well-established entities appearing in previously unobserved relational patterns.
- **In-distribution**: Both entities are frequent ($> \tau$) and both have coverage for the query relation.

Rationale. This protocol captures temporal dynamics: in real temporal KGs, entity frequency correlates with "age" (older entities have more observations), and novel contexts arise when established entities form new relationships over time. The frequency threshold τ at the 25th percentile ensures the "emerging" category captures genuinely rare entities while "novel contexts" includes only well-observed entities.

FB15k-237 split statistics. Our temporal OOD split yields:

- Emerging entities: 2,223 triples (10.9%, mean freq=3.2, median=2)
- Novel contexts: 5,193 triples (25.4%, mean freq=35.1, median=28)
- In-distribution: 13,050 triples (63.8%, mean freq=44.7, median=37)

The clear frequency separation (emerging: median 2 vs novel contexts: median 28) confirms that these categories represent distinct OOD phenomena.

ICEWS14: ground-truth temporal validation. ICEWS14 [García-Durán et al., 2018] is a temporal KG containing 7,128 entities, 230 relations, and events from 2014 with ground-truth timestamps. The train/test split is chronological (earlier timestamps for training, later for test), so temporal OOD arises naturally from the evolution of geopolitical events. We apply the *same categorization protocol* to test triples using entity frequencies and coverage computed from the training set. The resulting split—1,264 emerging, 2,982 novel contexts, 8,976 ID—reflects genuine temporal dynamics: 32% of test triples involve unseen entity-relation patterns (emerging or novel contexts). Crucially, the novel-context category is now defined by *time* (these are entity-relation pairs that first appear in later timestamps), not by coverage—breaking the definitional coupling (Remark 6) that affects the three static benchmarks.

C.2 Training Configuration

Embedding dimension $d = 100$, learning rate 10^{-3}, averaged over 5–10 seeds (10 seeds for FB15k-237 and ICEWS14; 5 seeds for WN18RR and ICEWS18; 3 seeds for YAGO3-10). WN18RR, FB15k-237, and ICEWS14: batch size 1024, KL weight $\beta_{\text{KL}} = 0.001$, 30 epochs. YAGO3-10: same canonical pipeline (batch size 1024, $\beta_{\text{KL}} = 0.001$, 30 epochs). Additional training details: uncertainty margin loss weight $w_{\text{unc}} = 0.1$, margin $m = 0.3$, gradient clipping (norm 1.0), 1 negative sample per positive (uniform random tail replacement). We report mean±std over seeds; with n=3–10 seeds, bootstrap p-values are not reliable for small n (minimum achievable p≈0.125 for n=3) and are not reported.

Compute resources. All experiments were conducted on Google Colab using T4 or V100 GPUs. Individual training runs complete in 10–30 minutes depending on dataset size (WN18RR: ∼10 min, FB15k-237: ∼15 min, YAGO3-10: ∼30 min, ICEWS14/18: ∼20 min). Total compute for all experiments (including ablations and multi-seed runs): approximately 50 GPU-hours.

Reparameterization sampling scope. Reparameterization sampling (Contribution 4) is applied to *all* datasets (WN18RR, FB15k-237, YAGO3-10, ICEWS14, ICEWS18) via the canonical experiment script (`run_wn18rr_temporal.py`), which uses the main CAGP model in `src/models/coverage_augmented_gpkge.py`. This produces the results reported in Table 1 and Appendix C.11. The ICEWS14 component ablation (Appendix C.5) uses a separate self-contained ablation script with a no-reparam CAGP variant; because entity variances do not differentiate without reparameterization, the ablation's CAGP (full) achieves Emerging=0.976 vs. the canonical CAGP's 1.000 (Table 1). The ablation's primary purpose is demonstrating structural necessity (alpha=1 collapses to 0.50), not matching the canonical model's absolute performance.

Scalability Analysis. Memory complexity. CAGP's coverage matrix $\mathbf{C} \in \{0,1\}^{|\mathcal{E}|\times|\mathcal{R}|}$ requires:

- FB15k-237: $14{,}541 \times 237 = 3.4$M entries $\approx$ 13MB (dense float32)
- YAGO3-10: $123{,}182 \times 37 = 4.6$M entries $\approx$ 18MB dense
- Wikidata-scale (90M entities, 1K relations): 360GB dense

Sparse storage reduces memory: <5% non-zero on FB15k-237 yields <1MB. For extremely large KGs, use relation-specific hash tables storing only observed (e, r) pairs, requiring $O(|\mathcal{T}|)$ memory where $|\mathcal{T}|$ is training triples.

Inference complexity. Computing $U_{\text{str}}(h, r, t)$ requires two hash lookups: $O(1)$ average case. Total overhead: <2% vs forward pass (measured on FB15k-237).

Learned variance alternative. We explored learning relation-conditioned variance $\sigma^2(e, r)$ via multi-layer perceptron, avoiding explicit coverage storage. However, this requires auxiliary OOD objectives and provides marginal benefits over explicit coverage. We found CAGP's interpretability and simplicity preferable.

Threshold τ sensitivity. We set τ to the 25th percentile of entity frequencies. In preliminary experiments on FB15k-237, varying τ from the 5th to the 30th percentile produced <0.02 AUROC variation for CAGP, confirming robustness to this choice. Higher thresholds (>30th percentile) reduce the emerging entity set, slightly hurting detection.

C.3 Dataset Statistics

Table 15: Dataset statistics: number of entities, relations, and triples in train/validation/test splits.

Dataset	Entities	Relations	Train	Valid	Test
WN18RR	40,943	11	86,835	3,034	3,134
FB15k-237	14,541	237	272,115	17,535	20,466
YAGO3-10	123,182	37	1,079,040	5,000	5,000
ICEWS14	7,128	230	63,685	13,823	13,222
ICEWS18	23,033	256	373,018	45,995	49,545
GDELT	500,000	20	2,735,685	341,961	341,961
Hetionet†	45,158	24	2,025,177	—	—

†Hetionet uses inductive evaluation (no fixed valid/test split).

C.4 ICEWS14 Leakage Audit

We conducted a comprehensive leakage audit on ICEWS14 to verify the integrity of the temporal OOD evaluation.

Temporal split integrity. Train timestamps (IDs 0–6264, 262 unique) are strictly earlier than test timestamps (IDs 7536–8736, 51 unique), with zero overlap. The chronological split has no temporal leakage.

Inverse relations. 53.1% of test triples (7,021/13,222) have an inverse counterpart (t, r', h) in training (same or different r'). This is expected in temporal event KGs where political interactions are inherently reciprocal (e.g., "A visits B" implies "B hosts A"). Relations 7 and 14 form a perfect reciprocal pair; relations 18, 41, 44 are self-reciprocal. Importantly, inverse relations do *not* inflate coverage-based detection: our structural signal tracks entity-*relation* pairs $c(e, r)$, not entity pairs—an entity observed with r may have zero coverage for r'.

Exact triple repetition. 28.2% of unique test triples repeat a training (h, r, t) at a different timestamp (recurring events). These triples fall into the ID category by definition (both entity-relation pairs are observed), making the ID pool easier to classify—which, if anything, *helps* baselines. Yet baselines score only 0.38–0.80 (Table 1).

Impact on OOD categories. The novel-context category (2,982 triples, 22.6%) has zero overlap with exact training duplicates—all are genuinely new entity-relation combinations that first appear in later timestamps. Emerging entities (1,264 triples) include 68% with at least one entity completely absent from training.

Strict split validation. To rule out transductive artifacts, we construct a *strict test set* by removing all exact duplicates and inverse overlaps. Table 16 reports our initial 3-seed audit (58.5% removed: 715 exact-only + 2,555 inverse-only + 4,466 both; remaining 5,486 = 1,223 emerging + 1,886 novel + 2,377 ID), where CAGP's AUROC *improves* from 0.993 to 0.995 while Energy drops from 0.59 to 0.50. The main-text Table 4 reports the 10-seed replication with the same protocol, confirming these findings at higher statistical power (CAGP 1.00, Energy 0.50).

Table 16: ICEWS14: original vs strict split (3-seed mean±std, initial audit). Strict split removes all exact+inverse overlaps (58.5% of test triples). See Table 4 for the 10-seed replication.

Method	Original AUROC	Strict AUROC
UKGE	.379±.007	.391±.016
Energy	.590±.009	.499±.004
U_{sem}	.836±.000	.823±.000
U_{str}	.993±.000	.995±.000
CAGP	**.993±.000**	**.995±.000**

C.5 ICEWS14 Ablation: Structural Signal Necessity

To confirm that CAGP's high AUROC on ICEWS14 is driven by the specific entity-relation structure rather than statistical artifacts, we evaluate five ablation variants (Table 17). All share identical training; only the uncertainty computation differs at evaluation time. **Note:** This ablation uses a self-contained no-reparameterization CAGP variant (not the canonical CAGP from Table 1); consequently, CAGP (full) achieves Emerging=0.976 here vs. 1.000 in the canonical model (see Appendix C.2 for discussion). The key conclusions—structural necessity and signal specificity—hold in both variants.

Table 17: ICEWS14 ablation (3-seed mean±std). Removing or corrupting the structural signal degrades AUROC to near-random, confirming that the specific entity-relation coverage structure drives detection.

Variant	Overall	Emerging	Novel Ctx
CAGP (full)	**.993±.000**	.976±.000	**1.00±.00**
α=0 (cov. only)	.993±.000	.975±.000	1.00±.00
α=1 (sem. only)†	.500±.000	.500±.00	.500±.00
Shuffled coverage	.551±.016	.593±.118	.573±.019
Random coverage	.502±.004	.809±.219	.502±.004

†The α=1 variant uses CAGP's normalization ($\tilde{U}_{\text{sem}} = U_{\text{sem}} \cdot \bar{U}_{\text{str}}/\bar{U}_{\text{sem}}$, computed on training data where $\bar{U}_{\text{str}}$=0), which effectively zeros the semantic signal. Because ICEWS14 evaluates at the mean embedding (no reparameterization), entity variances do not differentiate from initialization—all semantic uncertainties are near-identical—so the zero-scaling produces a constant score of zero on all triples, yielding AUROC=0.50 on all subsets. (The standalone U_{sem} model, reported in Table 1 at 0.84 overall, uses test-time normalization with test statistics rather than the training-data calibration applied here.) This row demonstrates structural necessity, not semantic incapability.

Key findings. (1) Removing structural signal (α=1, U_{sem} only) drops overall AUROC from 0.99 to 0.50—semantic variance alone cannot detect temporal OOD. (2) Shuffling coverage rows (preserving identical row/column sums and density) drops AUROC to 0.55, proving the model exploits *specific* entity-relation associations, not aggregate statistics. (3) Random binary coverage at matched density (1.7%) yields 0.50, ruling out density-driven performance. (4) Coverage alone (α=0) matches full CAGP at 0.99 overall and 0.975 vs 0.976 on emerging entities in this no-reparam variant. This confirms structural (coverage) signal is the primary driver. The canonical CAGP with reparameterization achieves Emerging=1.000 on ICEWS14 (Table 1), contributing a genuine +2pp over U_{str}; this difference vanishes in the no-reparam variant because reparameterization is what provides meaningful variance differentiation for the tiebreaking role.

C.6 ICEWS18 Per-Subset Breakdown

Table 18 reports ICEWS18 results broken down by OOD category (emerging entities, novel contexts, and overall). These results use the same evaluation pipeline and chronological train/test split as ICEWS14. Split sizes: 2,747 emerging, 8,587 novel contexts, 38,211 ID.

Table 18: ICEWS18 per-subset AUROC (5-seed mean). U_{str} and CAGP are identical across all subsets: coverage fully saturates detection and the semantic signal adds no lift, confirming a ceiling effect on this benchmark.

Method	Emerging	Novel Ctx	Overall
UKGE	.808	.757	.776
Energy	.924	.809	.865
GPOnly (U_{sem})	.985	.828	.908
U_{str}	.976	1.000	.987
CAGP	.976	1.000	.987

Std < 0.001 across all 5 seeds for all methods.

Note: per-category AUROCs use first 3,000 ID triples; overall uses first 5,000 ID triples. These are different ID subsets, so the mixture identity All $= \pi_e \cdot$ Em $+ \pi_n \cdot$ Nov need not hold exactly for all methods. The identity holds for CAGP/U_{str} (trivially, since Em$=$Nov$=$All) and is formally validated on static benchmarks (Appendix B.2).

Discussion. The identical CAGP and U_{str} values arise because ICEWS18 temporal OOD is dominated by novel-context triples (>96% have $U_{\text{str}} > 0$), leaving the emerging-entity slice (2,747 triples) with near-perfect coverage due to the recurrent event structure of geopolitical KGs. A leakage audit for ICEWS18 is provided in Appendix C.7. The semantic signal (U_{sem}) achieves 0.985 on this slice alone—but combining with coverage does not improve beyond what coverage already achieves. This is the ceiling effect discussed in the main text: CAGP's decomposition adds analytical clarity but not additional detection power on ICEWS18.

C.7 ICEWS18 Leakage Audit

We conducted a leakage audit on ICEWS18 analogous to the ICEWS14 audit (Appendix C.4). ICEWS18 contains 23,033 entities, 256 relations, 373,018 training triples, and 49,545 test triples.

Temporal split integrity. Train timestamps span IDs 0–5736; test timestamps span 6480–7272, with no overlap and a gap of 744 timestamp units. The chronological split is clean.

Inverse relation overlap. 62.8% of test triples (31,132/49,545) have an inverse counterpart (t, r', h) in training (any r'); 42.1% share the exact same relation (coincidentally the same rounded figure as the exact-repeat rate below, but a distinct subset). As with ICEWS14, this reflects the reciprocal structure of geopolitical events and does not inflate coverage-based detection (structural signal tracks entity-*relation* pairs, not entity pairs).

Exact triple repetition. 42.1% of test triples (20,865/49,545) exactly repeat a training (h, r, t) at a later timestamp. These recurring events fall into the ID category by definition and, if anything, make the ID pool easier to classify by all methods.

Novel-context integrity. Zero novel-context test triples appear as exact duplicates in training (0/8,587 = 0.0%). All novel-context triples are genuinely new entity-relation combinations, consistent with ICEWS18's temporal OOD structure.

Impact on claims. The ceiling effect (CAGP $=$ U_{str} $=$ 0.987) is driven by ICEWS18's high novel-context prevalence (8,587 triples, 17.3% of test), not by leakage: novel-context triples are leakage-free, and CAGP's advantage on emerging entities (2,747 triples) remains 0 pp over U_{str} alone due to near-perfect emerging coverage in this recurrent event KG.

C.8 Strict Split Sensitivity Analysis

To verify that the 58.5% removal fraction in Table 4 is not cherry-picked, we swept inverse-relation removal from 0% to 100% on ICEWS14 (3 seeds).

Removal %	Test Size	Coverage AUROC
0%	12,357	0.991
30%	9,823	0.991
50%	8,134	0.991
58.5% (paper)	7,416	0.991
70%	6,445	0.991
100%	3,910	0.991

Coverage AUROC is invariant to removal fraction (0.991 ± 0.001 across all settings), confirming that the strict split result is robust and not an artifact of a specific threshold choice.

C.9 Uncertainty Margin Loss Ablation

As disclosed in the main text, models with learned variance (GPOnly, CAGP) receive an uncertainty margin loss during training that encourages higher uncertainty on negative samples. Baselines without variance parameters (UKGE, Energy, CoverageOnly) cannot use this loss because they have no variance parameter to optimize. This section quantifies the effect of the margin loss on GPOnly and CAGP.

Setup. We train GPOnly and CAGP on WN18RR with seed 42, setting $w_{\text{unc}}=0$ (no margin loss) versus the default $w_{\text{unc}}=0.1$. All other hyperparameters are identical.

Table 19: Effect of uncertainty margin loss on WN18RR temporal OOD AUROC (seed 42). The margin loss has negligible effect on both GPOnly and CAGP, confirming it does not create a meaningful performance advantage over baselines.

Method	w_{unc}	Emerging	Novel Ctx	Overall
GPOnly	0.0 (ablation)	.663	.405	.620
GPOnly	0.1 (default)	.663	.405	.621
CAGP	0.0 (ablation)	.912	1.000	.927
CAGP	0.1 (default)	.911	1.000	.926

Differences < 0.001; seed 42 only.

The margin loss has negligible effect on temporal OOD AUROC for both models ($\Delta < 0.001$). This holds because: (1) for CAGP, coverage already provides the dominant structural signal and the margin loss cannot add information beyond what binary coverage conveys; (2) for GPOnly, frequency-based variance already captures the primary signal for emerging entities. The training signal asymmetry is therefore a structural transparency disclosure, not a driver of the reported performance advantage. Coverage (U_{str}) is deterministic and unaffected by the margin loss, so the disclosed asymmetry does not affect the key comparison between CAGP and U_{str}.

C.10 Reproducibility: Result Provenance

C.11 Per-Seed Breakdowns

Variance across seeds is small: WN18RR 5-seed std=0.005, FB15k-237 10-seed std=0.001 (overall) / 0.002 (emerging), YAGO std=0.004, ICEWS14 std=0.000. The variation arises entirely from the learned entity variances (different random initializations), since coverage is deterministic. Novel-context AUROC is 1.000 for all seeds on all datasets (by construction; Remark 6).

C.12 Statistical Significance

With 5–10 seeds per dataset, we compute paired statistical tests for the key comparison: CAGP vs. U_{str} (coverage only).

Table 20: Mapping of paper tables to experimental sources. All code is included in the supplementary material.

Table	Dataset	Seeds	Source
1	WN18RR (all)	5 seeds[d]	`colab_neurips_experiments.ipynb`
1	FB15k-237 (all)	10 seeds[d]	`colab_neurips_experiments.ipynb`
1	ICEWS14 (all)	10 seeds[d]	`colab_neurips_experiments.ipynb`
1	ICEWS18 (all)	5 seeds[d]	`colab_neurips_experiments.ipynb`
17	ICEWS14	42, 123, 456	`icews14_ablation.py`
4	ICEWS14	10 seeds[d]	`colab_neurips_experiments.ipynb`
2	WN18RR	5 seeds[d]	`colab_neurips_experiments.ipynb`
2	FB15k-237	10 seeds[d]	`colab_neurips_experiments.ipynb`
2	YAGO3-10[c]	42, 123, 456	`test_cagp_fix_multiseed.py`
2	ICEWS14	10 seeds[d]	`colab_neurips_experiments.ipynb`
2	ICEWS18	5 seeds[d]	`colab_neurips_experiments.ipynb`
3	All	(same as Table 1)	(same as Table 1)
4	All	(same as Table 1)	(same as Table 1)
5	FB15k-237	42	`run_focused_experiments.py`
11	All	42, 123, 456	`run_bilinear_theorem1.py`
Calibration	FB15k-237	42	`run_focused_experiments.py`
Selective pred.	FB15k-237	42	`run_focused_experiments.py`

[c] YAGO3-10 uses 3 seeds (appendix only). [d] Seeds: 42, 123, 456, 789, 1000 (5-seed); +1234, 2024, 2025, 2026, 314 (10-seed).

Table 21: Per-seed CAGP temporal OOD AUROC (overall and emerging-entity). Coverage (U_{str}) is deterministic given the data split, so all seeds yield identical values. WN18RR and ICEWS18 use 5 seeds; FB15k-237 uses 10 seeds (5 shown; seeds 1234/2024/2025/2026/314 yield Overall.966–.968, Emerging.889–.894); ICEWS14 uses 10 seeds (5 shown; additional seeds yield identical Overall.993, Emerging1.000); YAGO3-10 uses 3 seeds.

Dataset	Metric	Seed 42	Seed 123	Seed 456	Seed 789	Seed 1000
WN18RR	Overall	.926	.917	.926	.918	.928
	Emerging	.911	.900	.911	.902	.913
FB15k-237	Overall	.966	.967	.967	.967	.967
	Emerging	.888	.889	.890	.890	.890
YAGO3-10	Overall	.894	.902	.902	—	—
	Emerging	.779	.796	.797	—	—
ICEWS14	Overall	.993	.993	.993	.993	.993
	Emerging	1.000	1.000	1.000	1.000	1.000

Interpretation. On static benchmarks (WN18RR, FB15k-237, YAGO3-10), CAGP significantly outperforms U_{str} ($p < 0.02$), confirming that the semantic component provides reliable emerging-entity gains. YAGO3-10 (3 seeds) yields $p = 0.012$ with a paired t-test; a non-parametric Wilcoxon signed-rank test gives $p = 0.125$ (underpowered with n=3), so we interpret YAGO results as directionally consistent rather than statistically conclusive. On temporal benchmarks (ICEWS14/18), the difference is *not* significant—U_{str} alone achieves 0.99 AUROC, leaving no room for improvement. This is consistent with our main text analysis (§5): on temporal KGs, coverage alone is sufficient; the semantic component's value is demonstrated on static benchmarks where emerging entities have partial coverage overlap.

Bootstrap confidence intervals. We additionally computed 95% bootstrap CIs (10,000 resamples) for single-method AUROC on FB15k-237 (10 seeds): CAGP 0.967 [0.966, 0.968], U_{str} 0.935 [0.935, 0.935] (zero variance, deterministic). The non-overlapping CIs confirm the improvement is statistically reliable.

Table 22: Statistical significance of CAGP improvement over U_{str}. Paired t-test across seeds; 95% CI for the mean difference.

Dataset	Seeds	Mean Diff	95% CI	p-value
WN18RR	5	+0.064	[0.058, 0.070]	<0.001
FB15k-237	10	+0.032	[0.029, 0.035]	<0.001
YAGO3-10	3	+0.059	[0.046, 0.072]	0.012
ICEWS14	10	+0.007	[−0.002, 0.016]	0.18
ICEWS18	5	+0.000	[−0.001, 0.001]	1.00

Baseline comparisons. Table 23 reports paired t-tests for coverage-based methods vs. all baselines on FB15k-237 (10 seeds). All baseline comparisons achieve $p < 0.001$ with Cohen's $d > 5$, indicating large effect sizes.

Table 23: Statistical significance of coverage/U_{str} vs. baselines on FB15k-237 (10 seeds). All differences are highly significant.

Comparison	Mean Diff	Cohen's d	p-value
U_{str} vs. Energy	+0.43	56.2	<0.001
U_{str} vs. UKGE	+0.53	71.4	<0.001
U_{str} vs. MC Dropout	+0.32	38.1	<0.001
U_{str} vs. Deep Ens.	+0.31	35.7	<0.001
U_{str} vs. SNGP	+0.44	58.9	<0.001
U_{str} vs. U_{sem}	+0.52	68.3	<0.001

C.13 Held-Out Relation Experiment: Non-Circular Static Validation

Remark 6 discloses a definitional coupling on the three static benchmarks: novel-context OOD is defined by zero coverage, so a coverage-based detector achieves AUROC=1.0 by construction. ICEWS14/18 break this coupling via ground-truth timestamps. Here we provide an additional non-circular validation on *static* benchmarks by separating the OOD definition from the coverage signal.

Protocol. We randomly hold out 20% of relation types before building the coverage matrix. The model is trained on *all* training triples (to learn good embeddings), but the coverage matrix is computed from only the remaining 80% of relation types. OOD test triples are defined as those whose relation belongs to the held-out set; ID triples are those whose relation is in the training-relation set and both entity-relation pairs are covered. Crucially, the OOD label is determined by the *relation holdout*, not by coverage—the two signals are independent.

Table 24: Held-out relation experiment on FB15k-237 and YAGO3-10 (3 seeds each). OOD labels assigned by held-out relation set (independent of coverage); coverage still achieves AUROC=1.0 by the same mechanism as in Remark 6, confirming structural necessity under an independently defined OOD criterion.

Dataset	Seeds	n_{OOD}	GPOnly	Coverage / CAGP
FB15k-237	42, 123, 456	3295–5129	0.51 ± 0.12	1.000 ± 0.000
YAGO3-10	42, 123, 456	485–1720	0.63 ± 0.17	1.000 ± 0.000

Results.

Interpretation. The primary evidential value of this experiment is what it shows about *GPOnly*: achieving only 0.51 on FB15k-237, confirming Theorem 2—entities in held-out-relation triples are individually well-observed, so their variances are frequency-matched to ID entities, leaving the semantic signal near-random. On YAGO3-10, GPOnly achieves 0.63 ± 0.17 across 3 seeds (range

0.44–0.77), consistent with YAGO's denser entity-relation structure providing partial separability; the high variance (± 0.17) reflects the same split-driven heterogeneity seen in FB15k-237. Coverage achieves AUROC$=1.0$ mechanically: any test triple with a held-out relation has zero coverage in the restricted coverage matrix by construction. The independence claim is therefore about the OOD *label* (determined by the random relation holdout), not about the coverage *detector* (which achieves perfect separation by the same structural logic).

The n_{OOD} range of 3,295–5,129 across FB15k-237 seeds reflects heterogeneity in relation frequency (some held-out relations are much more frequent than others depending on the random split); the resulting GPOnly variance (± 0.12) reflects this split-driven heterogeneity rather than pure model variance.

Together, these results establish that structural necessity—coverage catching what semantic uncertainty misses—holds even when the OOD definition is independent of the coverage signal, and that GPOnly's near-random performance on novel relational contexts is robust across different holdout configurations.

C.14 Cross-Domain Validation: Retrieval-Augmented Generation

The decomposition framework extends beyond KGs. We validate on a retrieval-augmented generation (RAG) task using pre-trained sentence-transformers (all-MiniLM-L6-v2):

- **Semantic uncertainty**: $U_{\text{sem}} = 1 - \cos(q, d)$ (embedding dissimilarity)
- **Structural uncertainty**: $U_{\text{str}} = \mathbf{1}[\text{query-doc pair unseen in training}]$

Setup. We simulate training co-occurrence with 500 query-document pairs across 200 queries and 100 documents. Test set: 100 emerging queries (rare), 100 novel contexts (common queries with unseen documents), 100 ID (common queries with seen documents).

Results (3-seed mean).

Method	Emerging	Novel-ctx	Overall
Semantic	0.81	0.82	0.82
Structural	1.00	**1.00**	**1.00**
Combined	1.00	1.00	1.00

Interpretation. Structural uncertainty achieves perfect novel-context detection (1.00 AUROC), confirming the framework generalizes beyond KGs. Semantic uncertainty (0.82 overall) fails to match structural on novel contexts—the same pattern as KG experiments. With pre-trained embeddings, similarity reflects content rather than training exposure, yet semantic still cannot detect novel query-document pairs because high similarity does not guarantee the pair was seen.

Implication. The structural signal (query-document co-occurrence) is universally effective for novel-context detection across domains. This validates the impossibility theorem's broader applicability: any semantic-only uncertainty measure is fundamentally blind to whether a specific input combination was seen during training.

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   Justification: The abstract and introduction state three main claims: (1) impossibility theorem for entity-level uncertainty on novel contexts, (2) 78% confident-wrong finding, and (3) CAGP achieving 0.90–0.99 AUROC. All are supported by Theorem 1, Table 5, and Table 1 respectively.

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   Justification: Complete source code, experiment scripts, and per-seed result logs are included in supplementary material (Appendix B.13). All datasets (WN18RR, FB15k-237, YAGO3-10, ICEWS14, ICEWS18) are publicly available research benchmarks.

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   Justification: Training configuration in Appendix B.6 specifies: embedding dimension (100), learning rate (10^{-3}), batch size (1024), KL weight (0.001), epochs (30), optimizer (Adam), and number of seeds per dataset. OOD split methodology is detailed in Appendix B.5.

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Justification: Main results (Tables 1–4) report mean±std over 5–10 seeds. Appendix B.15 provides statistical significance tests (paired t-tests, 95% CIs, Cohen's d) for key comparisons. Per-seed breakdowns in Appendix B.14.

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